Raw EEG recording

TUSZ: 19 referential electrodes · CHB-MIT: 22 bipolar channels

LinOSS branch (OscSeq)

206,529 parameters for N = 19

Tokens: log|FFT| of each 1 s, 1 Hz bins 0–99, z-scored

x: (B, L, N, 100); one 19 × 100 token per second (TUSZ)

Linear 100 → 32 + electrode embedding

(B, L, N, 32)

1 BiLinOSS block per electrode, P = 32, shared weights

(B·N, L, 32)

Pool [mean ‖ max] over electrodes → Linear 64 → 96

(B, L, 96)

2 BiLinOSS blocks over the whole head, P = 96

(B, L, 96) = per-second summary h_t

Head: Linear 96 → 1

per-second logit ℓ_t: (B, L)

Blend 240 s windows every 120 s (sin² weights)

p_L(t), z_L(t) = logit p_L(t)

CatBoost branch

committee of 50 gradient-boosted tree models

16 longitudinal bipolar chains, resampled to 200 Hz

formed from the 19 referential electrodes

Filters (zero-phase): 0.5–35 Hz band + 60 Hz notch

high-pass order 1 · low-pass order 4 · notch Q = 30

60 s windows, stride 15 s

(n_windows, 16, 12000)

588 features per window

47 per channel × 4 regions × 3 statistics + 24 PLV

±7-window context (±105 s), 11 statistics per feature

588 + 11 × 588 = 7,056 columns

Committees: base 20 (588) + full 20 + sens 10 (7,056)

p_C(window) = mean of the three committee means

Per-second track: mean of the 1–4 windows covering t

p_C(t), z_C(t) = logit p_C(t)

z_L

z_C

1 s stacker (lr_ctx): logistic regression

p(t) = σ(β · s([z_C, z_L, z_C·z_L, S15(z_L), S61(z_L)]) + b)

S15, S61: centred 15 s / 61 s moving means; chosen on dev by patient-grouped CV

1 s output: event rule

p(t) → moving average (k s) → threshold → merge gaps → minimum duration → seizure events

h_t: per-second LinOSS summaries (dashed)

12 s output (clips)

overlooker head on h_t of the 12 s of the clip

+ WindowStacker [z_C, z_L, z_C·z_L]; threshold = best dev F1

60 s output (windows every 15 s)

overlooker head on h_t of the 60 s of the window

+ WindowStacker; post-processing of Zabihi et al. → events

CatBoost window / clip logit z_C (dashed)

Solid: the 1 s path. Dashed: inputs of the 12 s and 60 s heads; one LinOSS backbone serves every unit.